

NKANYEZI HAPPY-ANGEL MTIMKULU

Software Engineering Student | Mathematics & Statistics Graduate

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PROFESSIONAL SUMMARY

Mathematics and Statistics graduate and Software Engineering student with hands-on experience building applications in Java and Python, from client-server systems to full-stack and cloud-based projects. Comfortable working across the stack, applying object-oriented design, data structures and algorithms to solve real problems, and collaborating in Git-based, CI/CD-driven team environments. Strong analytical and mathematical foundation, with a growing portfolio of independent and team projects spanning AI, web development, mobile safety tools and cloud infrastructure. Seeking a Graduate Software Engineer role to apply and grow these skills in a professional environment.

TECHNICAL SKILLS

Programming Languages: Java, Python, JavaScript, R, SQL

Web Development: HTML, CSS, JavaScript, REST APIs

Cloud & Serverless: AWS (Lambda, API Gateway, S3), Serverless Architecture

Software Engineering: Object-Oriented Programming, Data Structures, Algorithms, Testing, Debugging

Tools & Practices: Git, GitLab, GitHub, Maven, Docker, CI/CD

Other: JSON, Client-Server Architecture, Problem Solving, Analytical Thinking

EDUCATION

WeThinkCode_ 2025 – Present
Diploma in Software Engineering

University of Johannesburg 2021 – 2025
BSc in Mathematical Science — Mathematics and Statistics

Hoër Tegnies Skool Carel de Wet 2016 – 2020
National Senior Certificate

EXPERIENCE

Peer Tutor | WeThinkCode_ Nov 2025 – Present

- Assisted fellow students with programming concepts, algorithms and data structures.
- Provided individual and group support with coding assignments and software engineering projects.
- Helped students identify and debug programming errors, explaining complex concepts clearly and accessibly.
- Supported collaborative learning and problem-solving within the software engineering programme.

Mathematics Tutor | University of Johannesburg 2025

- Delivered individual and group tutoring sessions covering calculus, algebra and statistics.
- Simplified complex mathematical concepts to improve student understanding.
- Developed students' analytical and problem-solving abilities, adapting teaching approaches to different learning styles.

Examination Assistant | Gauteng Department of Education 2024

- Reviewed examination scripts to verify marking accuracy and completeness.
- Checked captured marks against original examination scripts.
- Maintained accuracy, confidentiality and attention to detail throughout the verification process.

PROJECTS

Sentinel AI — Infrastructure Theft & Vandalism Detection (Hackathon Project)

IoT | Python | AI/Data Analysis | Sensors

- Part of a technical team building a system to detect and respond to theft and vandalism of critical infrastructure in real time.
- Explored IoT sensor placement and data capture around vulnerable infrastructure sites.
- Contributed to the software system that processed sensor data and triggered automated alerts and responses.

- Investigated the use of AI and data analysis to identify suspicious activity patterns and support predictive insights.
- Collaborated under time pressure to design, build and present a working end-to-end solution; team won Best Technical Team Overall.

Robot World — Software Engineering Project

Java | JSON | Sockets | Maven | JUnit | Git

- Built a client-server robot simulation in Java using socket-based communication between multiple clients and a central server.
- Implemented core game mechanics including robot movement, turning, firing, repairing and reloading.
- Designed JSON-based message structures to handle communication between client and server.
- Applied object-oriented programming, data structures and software design principles throughout the build.
- Used Maven and JUnit for build automation and testing; collaborated with a team via Git and GitLab CI/CD.

Salon Search and Booking Platform

JavaScript | HTML/CSS | REST API | SQL

- Designed and built a web application allowing users to search for nearby salons and book appointments online.
- Implemented search and filtering functionality to help users find salons by service, location and availability.
- Developed a booking workflow connecting the front end to a backend data store for appointments and salon listings.
- Applied client-server principles and structured data handling to keep booking information accurate and up to date.

Ride Safety Check — Driver Verification Tool

Python | APIs | Data Handling

- Built a tool to help riders verify the identity and safety credentials of e-hailing drivers (e.g. Uber, Bolt) before a trip.
- Designed logic to cross-check driver and vehicle details against expected records to flag mismatches.
- Focused on a simple, fast user flow so riders could confirm driver details with minimal friction.
- Applied problem-solving and analytical thinking to translate a real-world safety concern into a technical solution.

Smart Download Organizer

Python | File System Automation | Scripting

- Developed a script/application that automatically sorts and organizes files in a downloads folder by type, date or custom rules.
- Implemented file-monitoring logic to detect new downloads and move them into the correct folders in real time.
- Applied clean, modular code design to make the organizing rules easy to configure and extend.

Serverless Web App

AWS Lambda | API Gateway | Serverless Architecture | JavaScript

- Designed and deployed a web application using a serverless architecture to remove the need for managed server infrastructure.
- Built and connected backend functions to handle requests, process data and return responses via API Gateway.
- Focused on scalability and cost-efficiency by leveraging pay-per-use, event-driven cloud services.

CERTIFICATIONS & COURSES

- GenAI Course for Software Engineers — WeThinkCode_
- SheCodes Basics — HTML, CSS & JavaScript
- Data Engineering Course — DataCamp
- Data Analysis Course — Coursera (currently completing)

LANGUAGES

English | Sesotho | isiZulu

REFERENCES

Available upon request